

Syed Hamid Ali

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Education

Savitribai Phule Pune University (SPPU) **Pune, MH, India**
M.Tech. Atmospheric Sciences, jointly with the Indian Institute of Tropical Meteorology (IITM) 2021

University of Kashmir (KU) **Srinagar, J&K, India**
B.E. Mechanical 2018

Professional Appointments

Graduate Research Assistant, NOAA PERiLS Project **Purdue University**
2022–2024

Field Campaign Experience: Deploying Weather Instruments during PERiLS, and DELTA

Research Assistant **IITM Pune**
Contributed to several peer-reviewed studies 2021–2022

Graduate Research Assistant **IITM Pune**
M. Tech Thesis: Radar Derived Quantitative Precipitation Estimation 2020–2021

Peer-reviewed Publications

1. Gangane, A., Priyadarshini, P., Pawar, S. D., Gopalakrishnan, V., **Syed, H. A.**, & Dhangar, J. (2024). Impact of decreasing land–sea horizontal pressure gradient on the lightning activity over western India. *International Journal of Climatology*, 1–14. <https://doi.org/10.1002/joc.8542>
2. Gangane, A., S. Pawar, P. Priyadarshini, **H. A. Syed**, and V. Gopalakrishnan. Falling Trend of Winter Lightning and Its Synoptic Background: Study over Western India. Manuscript 2023EA003425, *Earth and Space Science*, AGU. [Revision Submitted]
3. Ghosh, R., D. Mudiar, S. D. Pawar, M. A. Domkawale, **H. A. Syed**, A. Hazra, and V. Gopalakrishnan, 2024: Observation of a dramatic increase in the positive cloud-to-ground lightning in the Indian summer monsoon season. *Atmospheric Research*, 298, 107119, doi:10.1016/j.atmosres.2023.107119.

Conference Presentations

Oral

1. **2023** DSD Characteristics of the Tornadic QLCS of Mar. 30 2022. PERiLS Science Meeting, Memphis
2. **2023** DSD Characteristics and Evolution of the Leading Stratiform Region of a Tornadic QLCS during PERiLS-2022 IOP#2 (30 March 2022). AMS 40th Radar Conf.
3. **2023** An Overview of Purdue's Mobile Disdrometer Operations in PERiLS 2023, AMS 40th Radar Conf.
4. **2022** Seventh WMO International Workshop on Monsoons (IWM-7): PyScanCf Introduction
5. **2021** INTROMET: Hourly radar-based quantitative precipitation estimation [Link to abstract](#) P. 456

Poster

1. **Syed, H. A.**, D. T. Dawson, and K. N. Brunner, 2024: Microphysical and thermodynamic insights into tornadic quasilinear convective systems from the Perils Field Campaign. 31st Conference on Severe Local Storms, AMS. [Link](#)
2. Dawson, Daniel T., Qin Jiang, Jacob Andrew Bruss, Matthew Graber, Funing Li, **Hamid Ali Syed**, Faith Vendl, Quinn Wilson, Michael I. Biggerstaff, and Sean M. Waugh (2022): Overview of Purdue's Mobile Disdrometer Operations in PERiLS 2022. In: 30th Conference on Severe Local Storms, Birmingham, AL, 22-26 November 2022. AMS, pp. 1-6.
3. **Syed, H. A.**, I. Sayyed, M. C. R. Kalapureddy, and K. K. Grandhi, 2022: PyScanCf –The open source python based library for IMD Doppler weather ... International Workshop on Monsoons (IWM-7). [Link to Poster](#)
4. **Syed, H. A.** and Kalapureddy, M.C.R. 2021, Radar-Based Quantitative Precipitation Estimation in Western Ghats, International Conference on Clouds and Precipitation, IITM Pune
5. **Syed, H. A.**, and M. C. R. Kalapureddy, 2021: Characterization of hourly radar-based quantitative precipitation estimation. Midwest Student Conference on Atmospheric Research. [Link to Poster](#)

Skills

Software Development.....

PyScanCf: The library for handling IMD radar data ([Link to Library](#))

Xradar: Xarray based radar toolkit for reading and writing radar data ([Link to Library](#))

Radarx: Xarray based radar toolkit for visualizations and methods ([Link to Library](#))

Programming.....

Python [IPyParallel, Dask Parallel, Parallel netCDF], LaTeX, R, FORTRAN 90/95, MATLAB, C/C++

Visualization and Statistics: Matplotlib, Scikit-learn, Ferret, Climate Data Operators (CDO), Origin, Panoply, Ncview

Data Experience.....

Radar Volume Scan Datasets (vol, Cf-Radial, Rainbow, Sigmet, Grid), Disdrometer, Raingauge, MRR

Satellite Observations: TRMM, GPM (L2, L3, IMERG), INSAT3D, SRTM

Reanalysis & Model datasets: MERRA2, MRMS, NCAR Reanalysis, ERA5, ERSST-V5, WRF, GPCP, IMDAA, OISST,

Professional Skills.....

Weather Forecasting, Technical & scientific writing, Machine Learning Algorithms, API Development, Python Packaging, Technical Documentation (SPHINX), Product Management, Debugging & Troubleshooting, Statistical Modeling, Version Control, Cloud Computing, Supercomputing (HPC), Xarray, GitHub, Gitlab, BitBucket, Visual Studio Code, Jupyter, Tensorflow, PyCharm, Docker, MS Office Suite, Multiprocessing

Key Projects

- **Microphysics, Kinematics, and Thermodynamics of Quasilinear Convective Systems (2022, 2024)**
Graduate Research Assistant, NOAA PERiLS Project, Purdue University
 - Developed and tested a method for hydrometeor size sorting in quasilinear convective systems (QLCS).

- Data acquisition, processing, quality analysis of WSR-88D and mobile research radar data along with PIPS, Sticknets, and lightning mapper arrays.
 - Presented the different components of this project at three different conferences.
 - Collaborations: Dr. Robin Tanamachi.
- **Wildfire Prediction Using Linear Models (2023)**
Weather and Research Forecasting Class Project, Purdue University
 - Developed and evaluated a linear regression model to predict forest fire damage based on key environmental factors.
 - **Genesis and Intensification of Cyclone Tauktae in the Arabian Sea (2022)**
Research Assistant, IITM Pune
 - Analyzed the impact of high sea surface temperatures, tropical cyclone heat potential, and atmospheric waves, including the Madden-Julian Oscillation, Kelvin, and Rossby Wave on Cyclone Tauktae's development.
 - Investigated environmental factors contributing to the cyclone's rapid and unprecedented intensification.
 - **PyScanCf: Python Toolkit for IMD Weather Radars (2021, 2022)**
Research Assistant, IITM Pune
 - Developed PyScanCf, a Python library for processing and visualizing IMD radar data.
 - Streamlined conversion of radar data to Cf-Radial and Cartesian formats for compatibility with tools like Py-ART and Wradlib. Enhanced data interpretation by enabling visualization of Max-CAPPI products.
 - **Radar-Based Quantitative Precipitation Estimation in the Western Ghats (2021)**
Graduate Research Assistant, IITM Pune
 - Conducted a study to develop new Z-R relationships tailored to the Western Ghats region for improved radar-based precipitation estimation.

Open Source Contributions

Xradar: <https://github.com/openradar/xradar>

PyDDA: <https://github.com/openradar/PyDDA/>

IMD Radar Network Dataset, 2023: <https://doi.org/10.6084/m9.figshare.22704910.v1>

Wradlib: <https://github.com/wradlib/wradlib>

Py-ART: <https://github.com/ARM-DOE/pyart>

DRpy: <https://github.com/dopplerchase/DRpy>

Professional Services

Member: American Meteorological Society (2022-present)

Reviewer: Journal of Open Source Software (2023-present)

Mentoring (undergrads)

Aug 2022 – May 2023: Faith Vendl

Extracurricular Courses/Certifications/Workshops

Dec 2020: Basics of Satellite Meteorology at Space Application Center, ISRO (Online)

Nov 2020: Machine Learning by Stanford University, Coursera (Audit)

Oct 2020: Understanding and Visualizing Data with Python, Coursera

Sep 2020: Data Analysis with Python, IBM, Coursera

Sep 2020: Python for Data Science, IBM, Coursera

Sep 2020: Python for Everybody by University of Michigan, Coursera

Feb 2020: From the Byte to Service: Trans-disciplinary Climate Research, IITM

Jan 2020: National Information System for Climate and Environment Studies and its Activities, NRSC ISRO & IITM

Jun 2020: Artificial Intelligence for Earth System Science Summer School, UCAR (remote)

Jul 2017: Course on Computer Concepts, National Institute of Electronics & Information Technology

May 2011: Course in C/C++, DOEACC

Comet/MetEd Courses, NCAR/UCAR ([Link](#))

Mar 2023: Gridded Forecast Verification and Bias Correction

Apr 2023: Analyzing and Evaluating Risk

Apr 2023: Impact-Based Forecasting: Identifying Hazards and Constructing Impacts Tables

Apr 2023: Met 101: Basic Weather Processes

Apr 2023: Met 101: Introduction to the Atmosphere

Apr 2023: NWP Essentials: NWP and Forecasting

Mar 2023: Communicating Forecast Uncertainty

Feb 2023: Communicating Probabilistic Forecasts

Awards

2014-2017 & 2019-2021: Merit cum means scholarship – Ministry of Minority Affairs, Govt. of India

Language Skills

Kashmiri, English, Urdu, Hindi